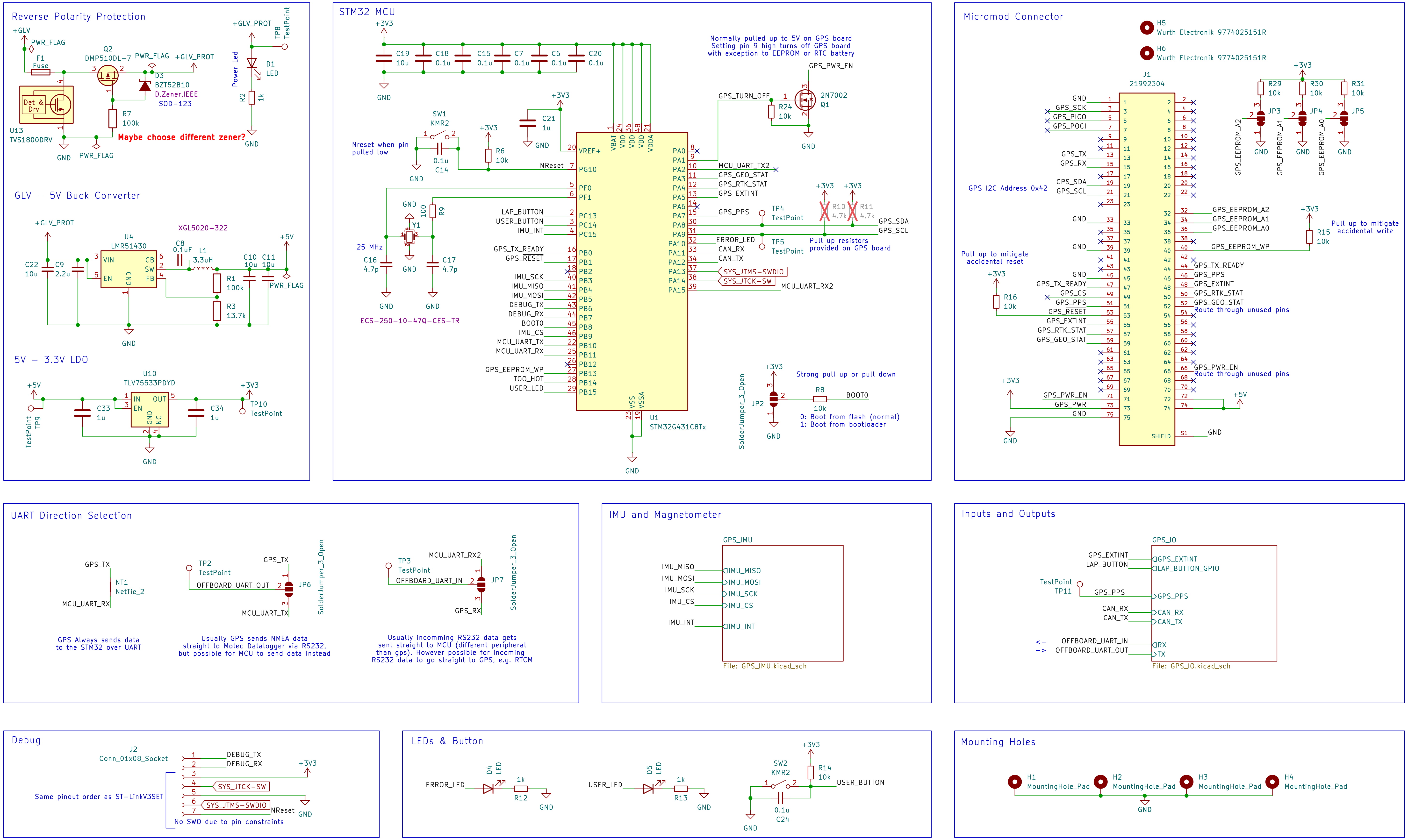
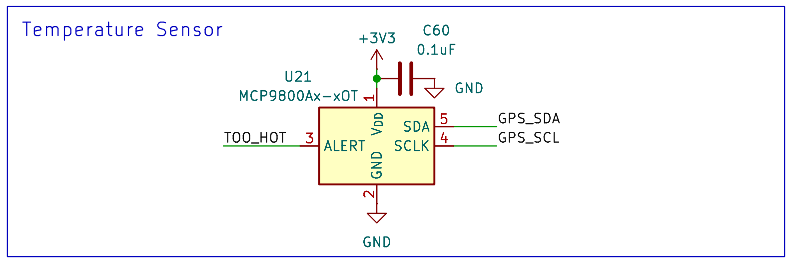


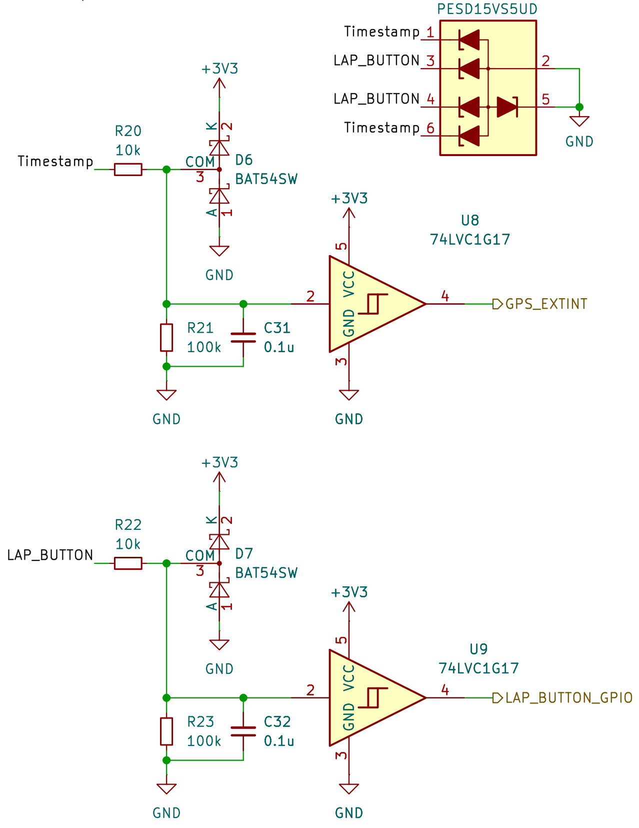


Vehicle: STAG 12
Drawn By: Adam Eastman
Checked By: Annabel Owen, Ray Wang
CAD Part:
SUFST
Sheet: Root
File: gps-mainboard.kicad_sch

Title: GPS Mainboard

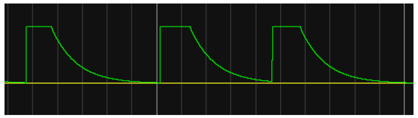
Size: A3 | Date: 2026-04-01 | **Rev: v1.0.0**
KiCad E.D.A. 9.0.4 | Id: 1/3

Driver Inputs



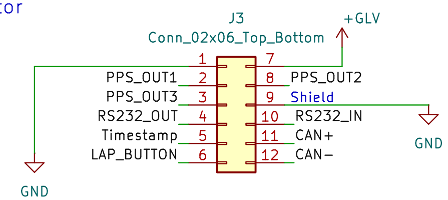
Signal conditioning of inputs.

Timestamp and Lap Reset inputs are capable of 24V
Current limiting resistor and BAT54SW bounds signal to 3.3V
RC filter to debounce. Tau=10ms
Schmitt trigger for precise turn on thresholds

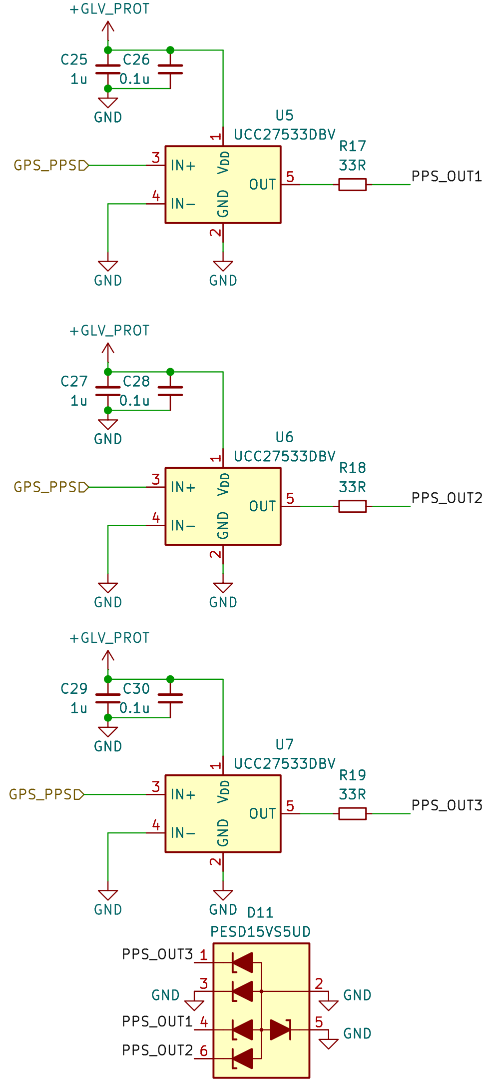


https://is.gd/Ufd4WO

Connector



PPS Signal Drivers

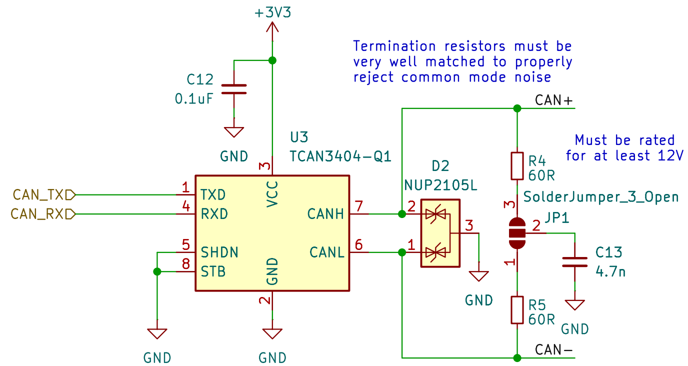


GPS sends a PPS signal every second
precise to the microsecond.

In order to distribute this signal
to devices in the car, gate drivers
are used, rather they are
used as line drivers.

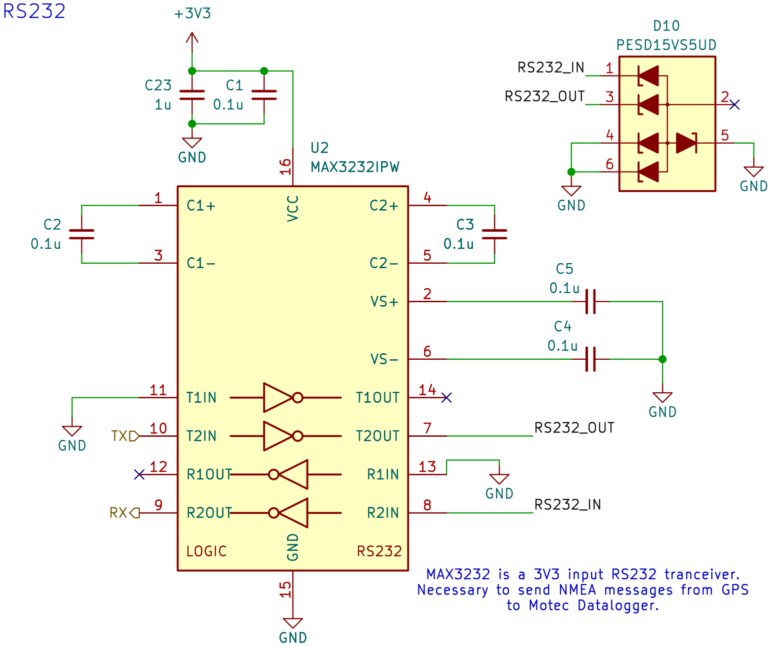
The gate drivers are capable of fast switching of GLV,
essentially level shifting the PPS signal,
making it more noise robust and capable of
driving many devices.

CAN Tranceiver



Shutdown and standby disabled
Always operating in normal mode

RS232



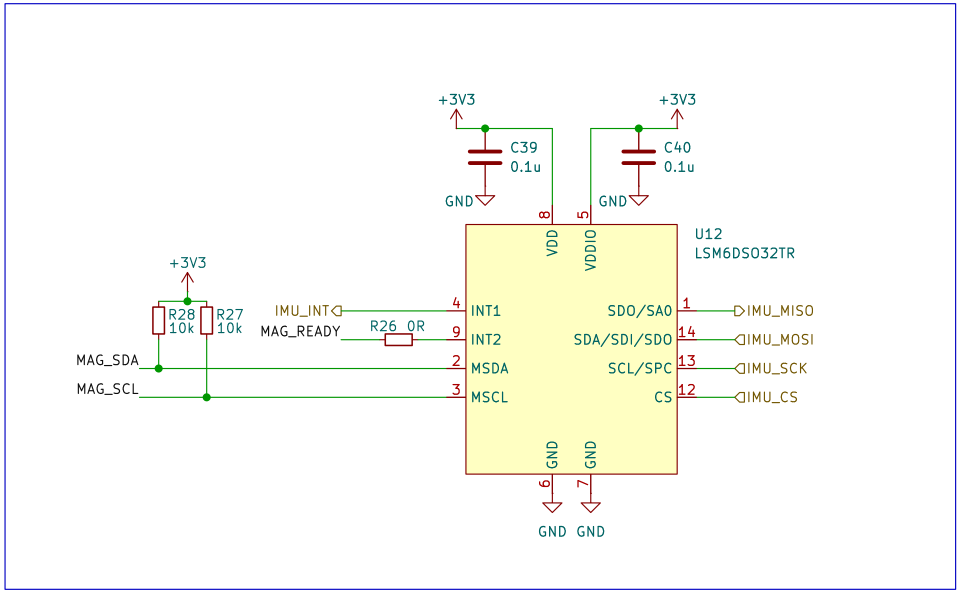
MAX3232 is a 3V3 input RS232 tranceiver.
Necessary to send NMEA messages from GPS
to Motec Datalogger.

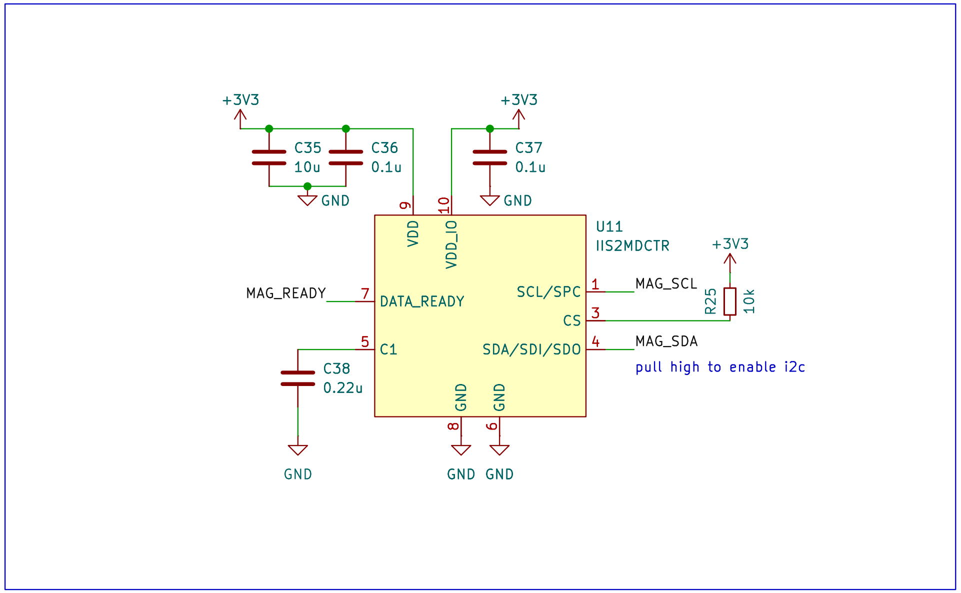
Vehicle: STAG 12
Drawn By: Adam Eastman
Checked By: Annabel Owen, Ray Wang
CAD Part:
SUFST

Sheet: GPS_IO
File: GPS_IO.kicad_sch

Title: GPS Mainboard

Size: A4	Date: 2026-04-01	Rev: v1.0.0
KiCad E.D.A. 9.0.4		Id: 2/3





Vehicle: STAG 12
Drawn By: Adam Eastman
Checked By: Annabel Owen, Ray Wang
CAD Part:
SUFST

Sheet: GPS_IMU
File: GPS_IMU.kicad_sch

Title: GPS Mainboard

Size: A4 | Date: 2026-04-03 | **Rev: v1.0.0**
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